

Despite their scale, architectural sophistication, and instruction tuning, SOTA LLMs exhibit the same qualitative behaviors as controlled models trained in DATAALCHEMY: strong in-distribution performance, sharp degradation under task, length, and format shifts, and limited benefit from increased model capacity. This alignment between synthetic and real-world settings supports the claim that the effectiveness and brittleness of CoT reasoning are fundamentally governed by distributional alignment, rather than by pretraining scale or architectural advances alone.

G. Additional Qualitative Analysis

G.1. Failures in Task Generalization

G.1.1. Failures caused by Transformation Generalization

The following case illustrates the failure caused by the transformation partial OOD setting discussed in Section C.1.1. Specifically, the model is trained exclusively on the compositional transformation $f_1 \circ f_2$, but is evaluated on the unseen composition $f_1 \circ f_1$.

```
Query:
H U S P [F1] [F1] <think>
Response:
U H F C [F2] <answer> H F C U
Ground-Truth:
U H F C [F1] <answer> H U S P
```

The results demonstrate that the model persists in applying the training composition ($f_1 \circ f_2$) rather than adapting to the new sequence ($f_1 \circ f_1$), even though the constituent transformation f_1 was observed during training. The model incorrectly generates the token [F2] in the reasoning step, mirroring the fixed pattern learned from the training data.

G.1.2. Failures caused by Element Generalization

The following case illustrates the failure caused by the element-level OOD setting discussed in Section C.1.2. Specifically, the training data was restricted to visible elements A–M, while the model was evaluated on queries consisting exclusively of elements N–Z.

```
Query:
N N N Q [F2] [F1] <think>
Response:
A A C E [F1] <answer> N N P R
Ground-Truth:
N N Q N [F1] <answer> A A D A
```

In the generated reasoning, the model incorrectly outputs AACE instead of the expected intermediate result NNQN. This error arises because all training queries consisted solely of elements A–M. Consequently, the model fails to generalize the transformation f_2 to the unseen elements; instead, it reverts to the training distribution, attempting to replicate the A–M patterns observed during training.

G.2. Failures in Length Generalization

G.2.1. Failures in Text Length Generalization

The following failure case demonstrates the model’s inability to generalize to unseen text lengths. In this experiment, the model was trained on sequences of length four but evaluated on sequences of length five.

```
Query:
I G L L Q [F1] [F2] <think>
Response:
T Y Y [F2] <answer> T Y Y V
Ground-Truth:
V T Y Y D [F2] <answer> T Y Y D V
```

Despite the increased length of the input query (five atoms), the model fails to adapt. As seen in the generated answer (T Y Y V), the model rigidly adheres to the length constraint observed during training, outputting a sequence of four atoms instead of the required five.

G.2.2. Failures in Reasoning Step Generalization

The following case shows that model trained under $f_1 \circ f_1$ tried to reproduce the length in training data by adding tokens in the reason chain even prompted with seen transformation f_1

```
Query:
A A B D [F1] <answer>
Response:
N O A Z N N O Q [F1] <answer> A A B D
Ground-Truth:
N N O Q
```

G.3. Failures in Format Generalization

The following failure case demonstrates the model’s inability to generalize under format change. Specifically, here we use delete as the format-changing mechanism.

```
Query:
A A A T [F1] <answer>
Response:
N G N G Y
Ground-Truth:
N N N G
```

H. Proof of Theorems

H.1. Proof of CoT Generalization Bound

Proof. Recall that the empirical and population training risks and the test risk are

$$\hat{R}_{\text{train}}(f_\theta) = \frac{1}{n} \sum_{i=1}^n \ell(f_\theta(x_i), y_i), \quad (18)$$

$$R_{\text{train}}(f_\theta) = \mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{train}}} [\ell(f_\theta(x), y)] \quad (19)$$

$$R_{\text{test}}(f_\theta) = \mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{test}}} [\ell(f_\theta(x), y)] \quad (20)$$

with $(x_i, y_i) \sim \mathcal{D}_{\text{train}}$ i.i.d., and the loss satisfies $0 \leq \ell(f_\theta(x), y) \leq B$ for all (x, y) . We also defined

$$\Delta(\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}}) := \text{TV}(\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}}). \quad (21)$$

We decompose

$$R_{\text{test}}(f_\theta) - \hat{R}_{\text{train}}(f_\theta) = (R_{\text{test}}(f_\theta) - R_{\text{train}}(f_\theta)) + (R_{\text{train}}(f_\theta) - \hat{R}_{\text{train}}(f_\theta)) \quad (22)$$

and bound the two terms separately.

First, let $P = \mathcal{D}_{\text{train}}$, $Q = \mathcal{D}_{\text{test}}$ and $g(x, y) = \ell(f_\theta(x), y)$. Using the fact that for any measurable g with $|g| \leq B$,

$$|\mathbb{E}_P[g] - \mathbb{E}_Q[g]| \leq 2B \text{TV}(P, Q) \quad (23)$$

we obtain

$$|R_{\text{test}}(f_\theta) - R_{\text{train}}(f_\theta)| \leq 2B \Delta(\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}}) \quad (24)$$

and in particular,

$$R_{\text{test}}(f_\theta) \leq R_{\text{train}}(f_\theta) + 2B \Delta(\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}}) \quad (25)$$

Second, define $Z_i := \ell(f_\theta(x_i), y_i) \in [0, B]$. Then

$$\hat{R}_{\text{train}}(f_\theta) = \frac{1}{n} \sum_{i=1}^n Z_i, \quad R_{\text{train}}(f_\theta) = \mathbb{E}[Z_i] \quad (26)$$

By Hoeffding's inequality, for any $\varepsilon > 0$,

$$\Pr\left(\hat{R}_{\text{train}}(f_\theta) - R_{\text{train}}(f_\theta) \geq \varepsilon\right) \leq \exp\left(-\frac{2n\varepsilon^2}{B^2}\right) \quad (27)$$

Setting the right-hand side to δ and solving for ε yields $\varepsilon = B\sqrt{\frac{\log(1/\delta)}{2n}}$, so with probability at least $1 - \delta$,

$$R_{\text{train}}(f_\theta) \leq \hat{R}_{\text{train}}(f_\theta) + B\sqrt{\frac{\log(1/\delta)}{2n}} \quad (28)$$

Combining Equation (24) and Equation (28), we conclude that, with probability at least $1 - \delta$,

$$\begin{aligned} R_{\text{test}}(f_\theta) &\leq R_{\text{train}}(f_\theta) + 2B \Delta(\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}}) \\ &\leq \hat{R}_{\text{train}}(f_\theta) + B\sqrt{\frac{\log(1/\delta)}{2n}} + 2B \Delta(\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}}) \end{aligned} \quad (29)$$

which completes the proof. $\square$